2026 TOYOTA INNOVATION CHALLENGE · COLLABORATIVE ROBOTICS

HANA

A collaborative robot that doesn't just avoid people —
it understands them, and works alongside them.

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Why are factory robots still in cages?

Traditional industrial arms are fast and precise, but have a glaring weakness. They are blind to people. However, fencing them off wastes floor space and makes real human-robot teamwork impossible.

On a real line, the robot and the worker should share one task, each handling the part they do best, embodying Toyota's Just-In-Time philosophy.

Our focus: a robot that knows where the worker is, and what they intend.

Built for TMMC — where 8,500 Team Members assemble the Lexus RX & RAV4 alongside automation every day.

ISOLATED

Robots run behind fences — no shared workspace with people.

REACTIVE ONLY

Standard cobots just halt when bumped — they don't cooperate.

OUR ANSWER

A robot aware of the worker's hands and intent, in real time.

Three layers

LAYER 1
Foundation

Vision-guided pick & place. The arm finds the part and moves it into a container reliably.

LAYER 2
Tiered Safety

Two zones around the robot: a Caution zone slows it, a Danger zone stops it.

LAYER 3
Handoff Protocol

Hand present -> deliver into the worker's hand. No hand -> place in the container.

DESIGN DECISION
Sensing the hand

- **Color / skin tone** *rejected — fails in changing light*
- **Motion tracking** *rejected — false triggers on movement*
- **MediaPipe hand model** *chosen — robust, low-light tolerant*

From pixels to motion

Orbbec camera -> **Homography** calibration -> **MediaPipe + HSV** detection -> **Zone & state** logic (Jidoka) -> **Dobot** Magician arm

1

12-point calibration

A homography matrix maps every camera pixel to a real (x,y) in the arm's frame — the gripper goes exactly where the part is.

2

Fingertip-level zones

All five fingertips on up to two hands are checked every frame; the most dangerous zone wins — conservative by design.

3

Runs at ~33 fps

Frames are downscaled and contrast-enhanced (CLAHE) before inference, keeping detection fast and stable for live control.

Safety first — the way Toyota means it

CLEAR

No hand detected

Robot runs at full speed.

CAUTION ZONE

Hand near the workspace

Robot slows to 50% as an early warning.

DANGER ZONE

Hand in the work area

Stops instantly; auto-resumes 3 s after the hand clears.

ALIGNED TO

Recognized safety practice

ISO/TS 15066

Speed & Separation Monitoring — the basis of our tiered zones.

CSA Z434

Canadian standard for industrial robot safety.

Fail-safe default

Any uncertainty -> the robot stops. Jidoka: it halts itself.

The 8,501st team member

A companion, not a replacement

TMMC runs on 8,500 valued Team Members. Hana joins them with the goal to assist, not displace.

Parts that arrive in your hand

The Handoff Protocol delivers a part the moment the worker reaches for it — Just-In-Time, at human scale.

It watches over the worker

An operator monitor flags drowsiness, micro-sleeps and yawns — safety for the human, not just from the robot.

It responds to gestures

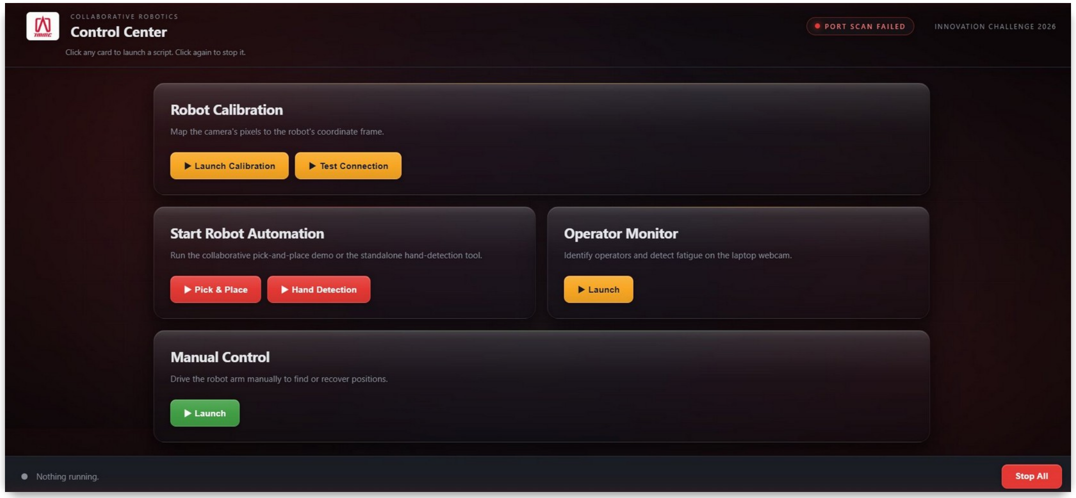
Flash a peace sign and Hana pauses and waves back — natural, hands-free interaction the worker controls.

Toyota’s core philosophies in our design

Jidoka *Automation with a human touch*	Hand in the danger zone -> the robot stops itself; a failed pick -> it pauses and retries.
Just-In-Time *The right part, exactly when needed*	The Handoff Protocol releases a part only when the worker reaches for it.
Andon *Visual status, at a glance*	The web Control Center + on-screen green / amber / red light show state instantly.
Kaizen *Continuous improvement*	Every run is logged and self-checked, so each cycle can be measured and improved.
Genchi Genbutsu *Go and see for yourself*	Hana sees the real workspace with its camera each cycle before acting — it never assumes.

What's running — and how we run it

- ✓ Vision-guided detection + auto tray (drop-zone) finding
- ✓ Picks every detected object, one after another
- ✓ Two-tier safety zones: slow in caution, stop in danger
- ✓ Handoff: delivers to the hand when present, tray when not
- ✓ Peace-sign gesture -> robot pauses and waves back
- ✓ Operator ID + fatigue monitor (drowsiness, yawns)



One Control Center launches & monitors every module

Live robot-connection status · one-click start/stop · TMMC-branded UI

Smart where it counts

WHERE THE COST IS

Robotic arm Dobot Magician — the single largest cost.

Depth camera Orbbec sensor for vision-guided detection.

Software Python, OpenCV, MediaPipe — $0 licensing.

HOW WE KEPT IT LOW

100% open-source stack

No license fees — the entire vision & control pipeline is free software.

Reused provided hardware

Built on the starter arm and camera; zero custom fabrication.

Software scales for free

The same code runs on a larger industrial arm — no rewrite needed.

From prototype to plant floor

IF WE HAD MORE TIME

- **Orientation-aware grip**
  Rotate the gripper to match a part's angle and grab irregular shapes.
- **Depth-based 3D zones**
  Move from 2D image regions to true 3D safety volumes.
- **Manual takeover mid-run**
  Let an operator seize control during automation at any moment.

SCALING TO INDUSTRY: RISKS & MITIGATION

Lighting variance
Already softened with CLAHE; depth sensing removes the rest.

Single-camera blind spots
Fuse multiple cameras for full workspace coverage.

Certification
Engineer to ISO/TS 15066 SSM for a certified deployment.

Toyota Motor Manufacturing Canada Inc.

People and machines, each doing what they do best.

TMMC spent 38 years building one of the world's best human–robot factories.

We built a robot that reads human intent. The principle is the same.

Thank you — David Wang · Justin Fang · Steven Qi · Heidi Wang

Electrical Engineering · Hana, the Collaborative Robot